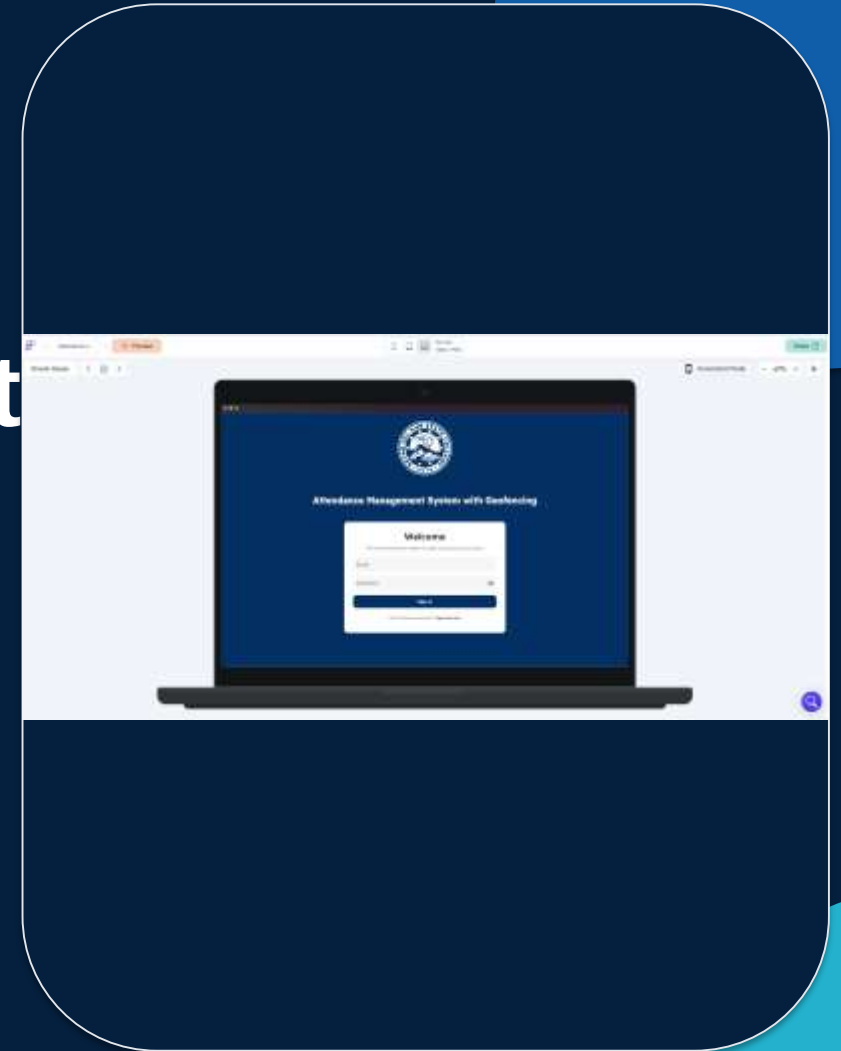




Mobile Biometrics Attendance Management System with Geofencing

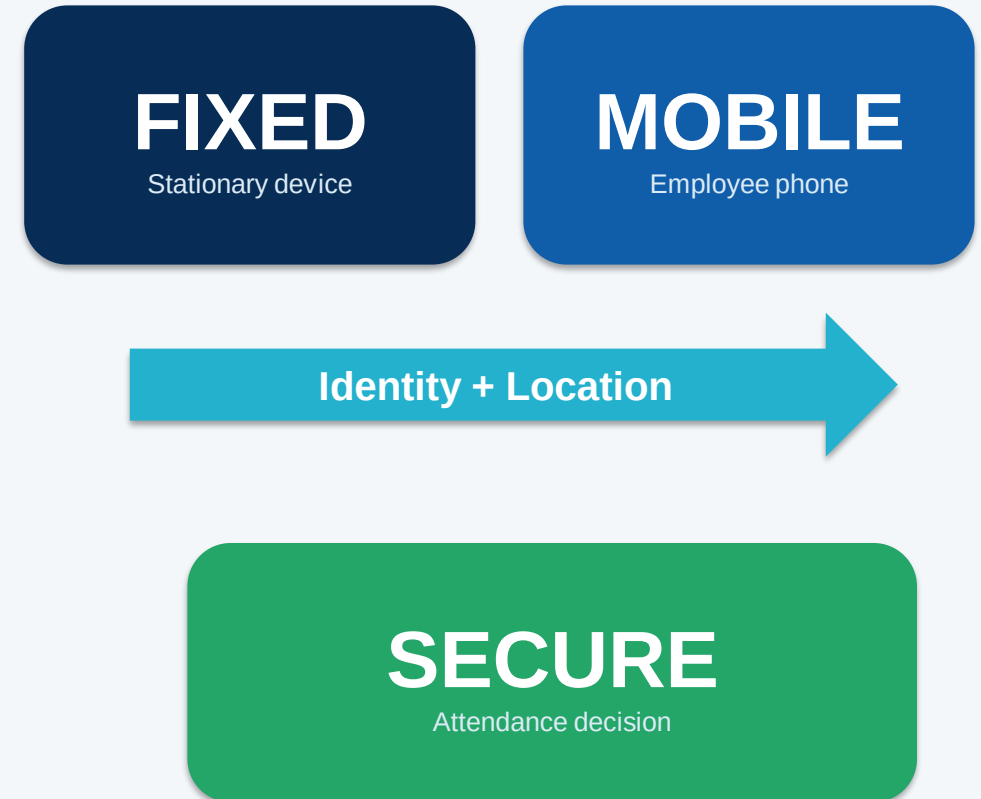
A secure, location-aware approach to employee attendance.



The Problem: Attendance Is More Than Identity

01

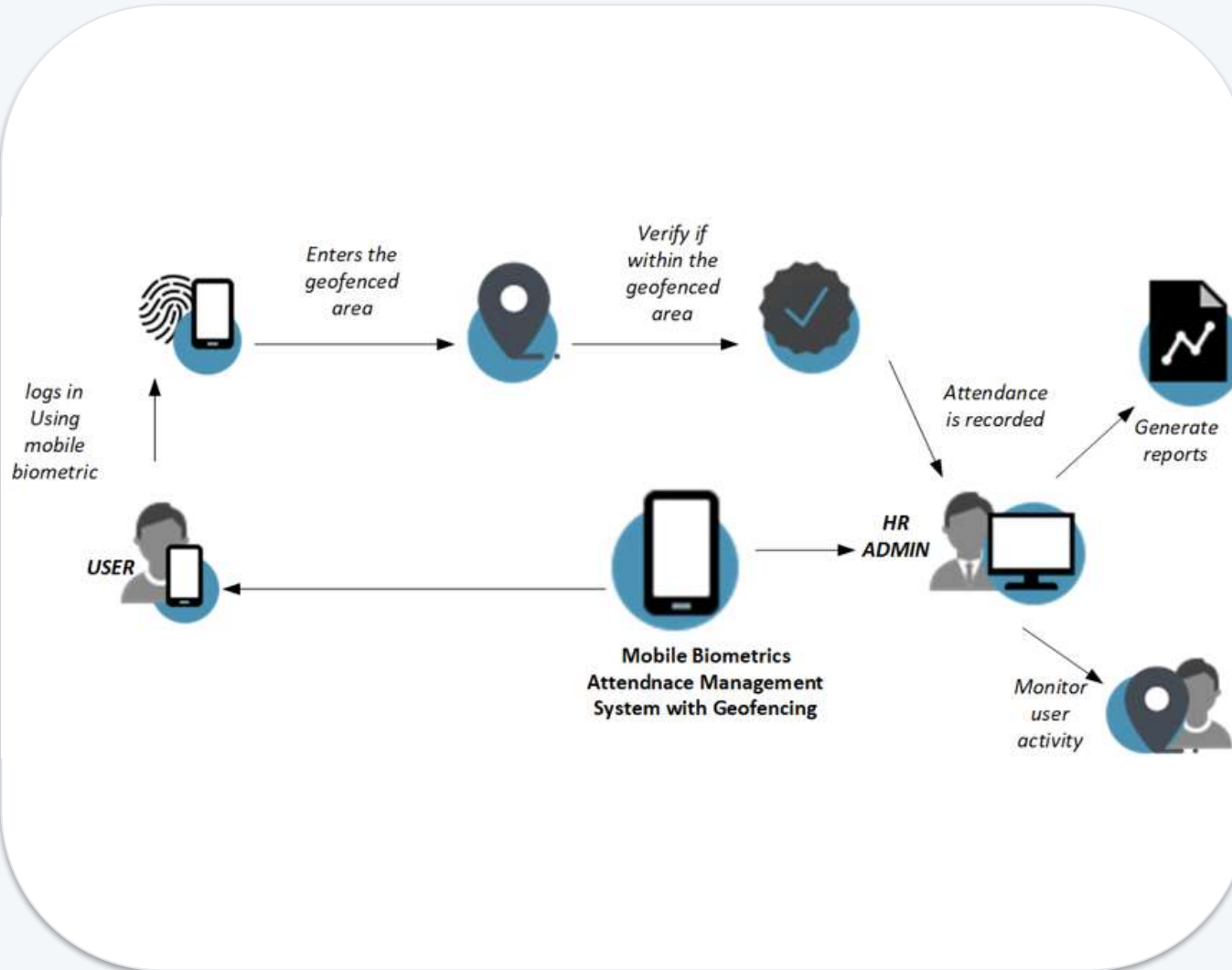
- Stationary fingerprint devices create congestion during peak hours.
- Attendance cannot be logged during blackouts, brownouts, or power surges.
- Traditional biometric-only setups verify identity, but not whether the employee is at an authorized location.
- The project replaces the fixed-device dependency with smartphone biometrics + GPS geofencing.



02

The Core Idea: Two Signals, One Attendance Decision

02



- Employee registers through the mobile platform.
- The phone checks the current GPS position.
- The system validates whether the user is inside the defined geofence.
- Only then is biometric attendance enabled and recorded.
- HR receives real-time records and reporting tools.

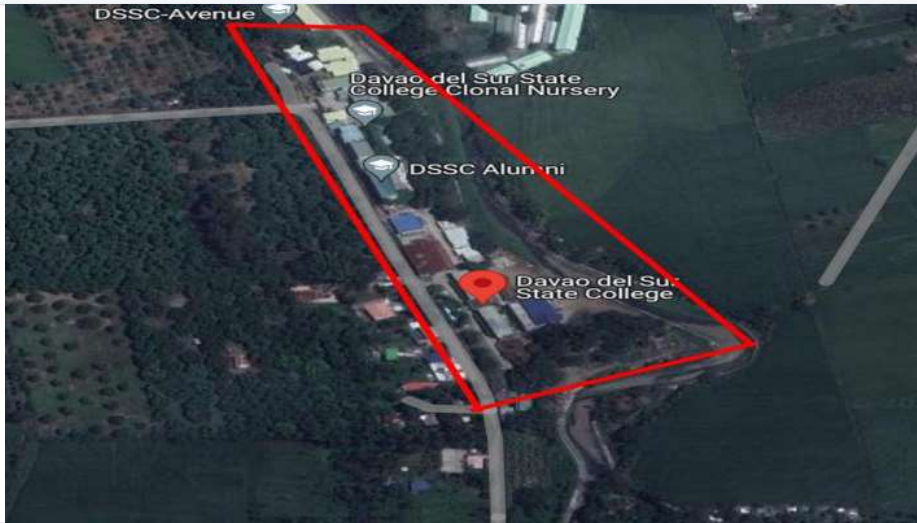
01
Identity

02
Location

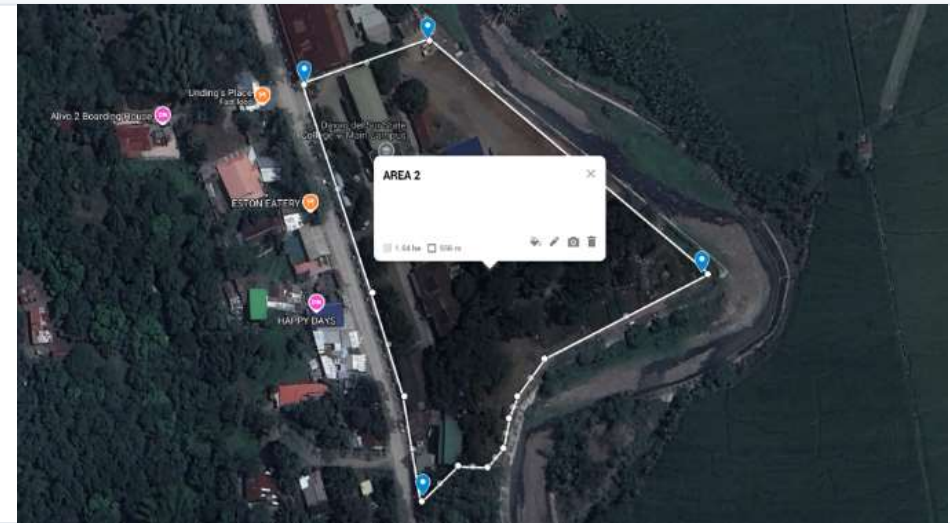
03

A Geofence Built From the Actual Campus

03



Area 1 • campus boundary



Area 2 • second permitted zone

MAP

Draw campus polygons

COORDINATES

Extract latitude/longitude

VALIDATE

Ray Casting algorithm

DECIDE

Allow / deny attendance

04

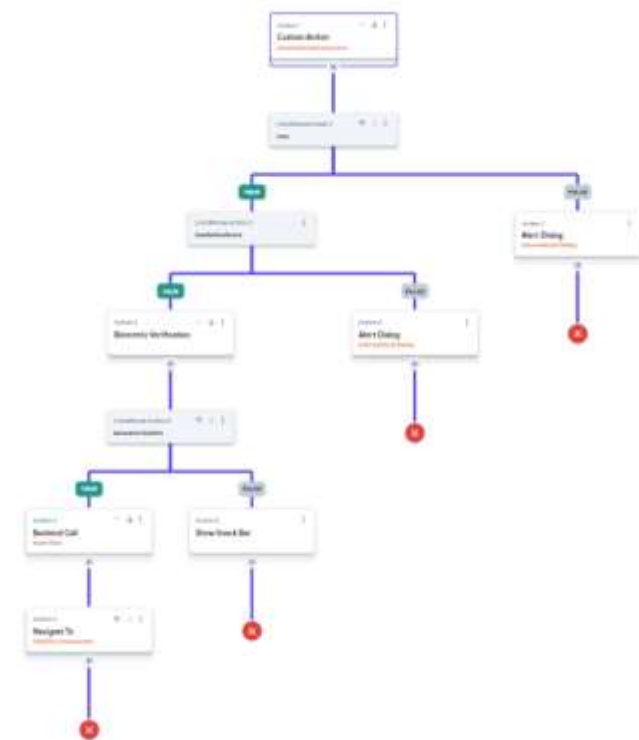
How the Geofencing Logic Works

04

```
// Check if location services are enabled
serviceEnabled = await Geolocator.isLocationServiceEnabled();
if (!serviceEnabled) {
  return 'Location services are disabled.';
}

// Request location permission if necessary
permission = await Geolocator.checkPermission();
if (permission == LocationPermission.denied) {
  permission = await Geolocator.requestPermission();
  if (permission == LocationPermission.denied) {
    return 'Location permissions are denied.';
  }
}

if (permission == LocationPermission.deniedForever) {
  return 'Location permissions are permanently denied.';
}
```

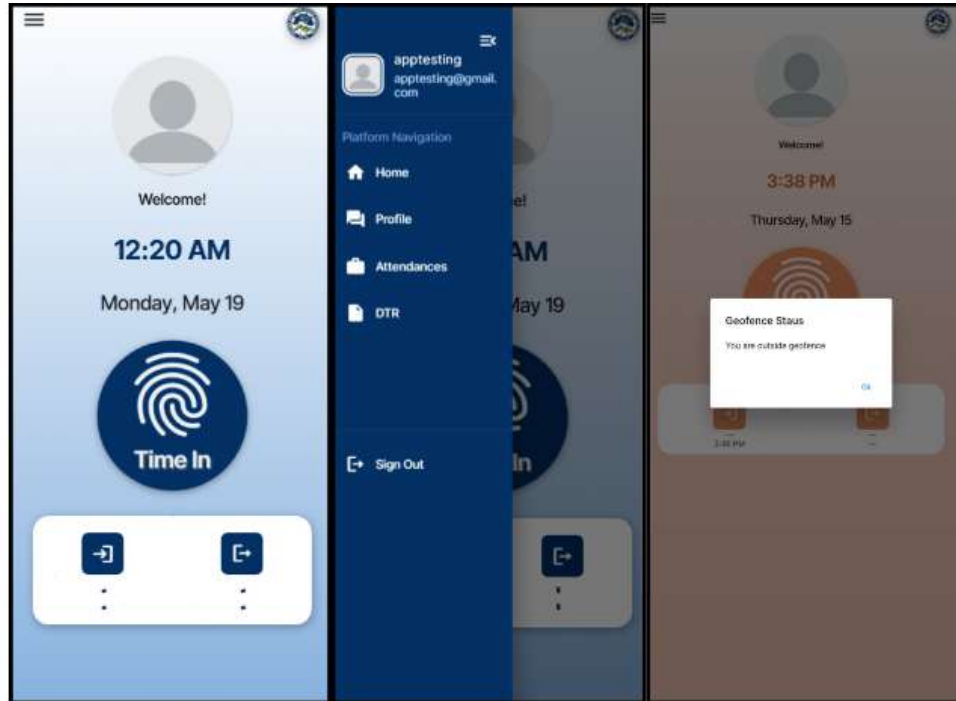


GPS location → polygon check → inside = biometric access • outside = attendance blocked

05

What the User Actually Experiences

05



(a)

(b)

(c)

Mobile attendance flow



Location restriction test

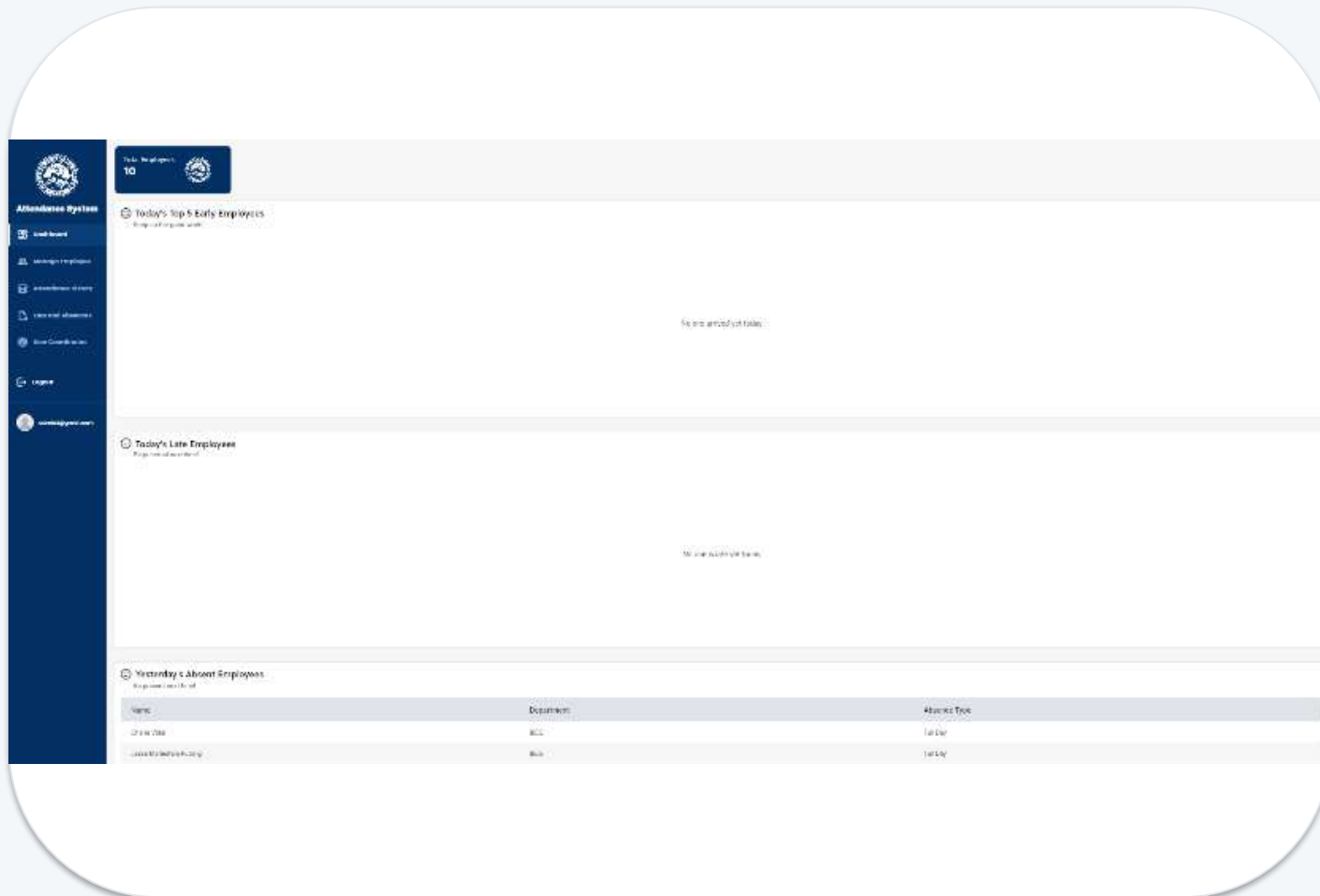


Campus test location

06

The HR Admin Side: From Logs to Decisions

06



- Dashboard overview of total employees.
- Top 5 early employees and today's late employees.
- Previous-day absences with department and absence type.
- Attendance history, late/absence monitoring, and user coordinates.
- Automated attendance reports for HR operations.

07

Validation: The System Correctly Controls Access

07

5 / 5

Geofence tests passed

100%

Correct allow / deny

2

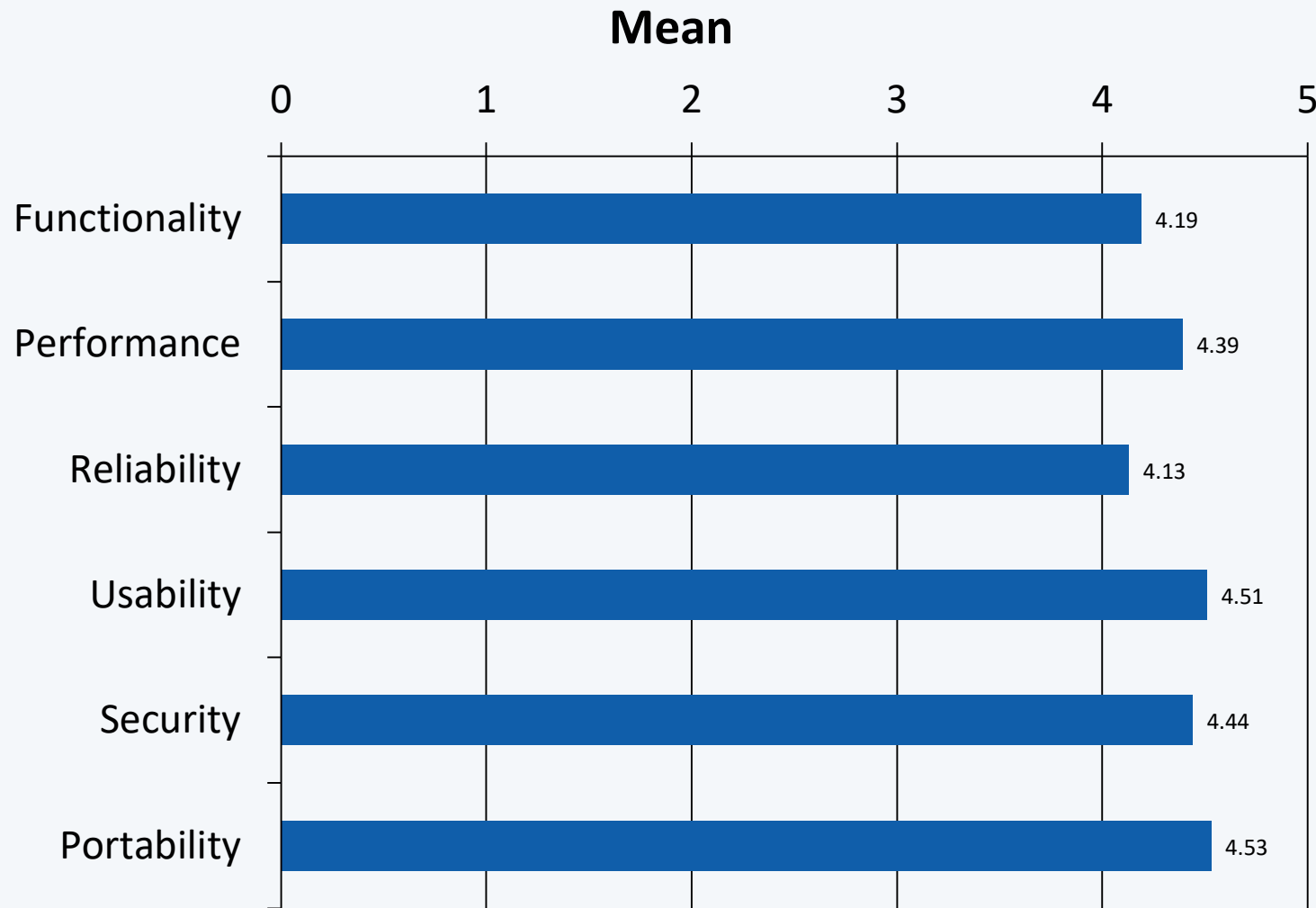
Defined campus areas

#	Test location	System response	
1	Inside • Gate 1	ALLOWED	✓
2	Inside • Gate 2 edge	ALLOWED	✓
3	Outside • Area 2	DENIED	✓
4	Outside • Area 1	DENIED	✓
5	Inside • Area 2	ALLOWED	✓

08

User Evaluation: Where the System Stands Out

08



4.53

Portability • highest

4.51

Usability

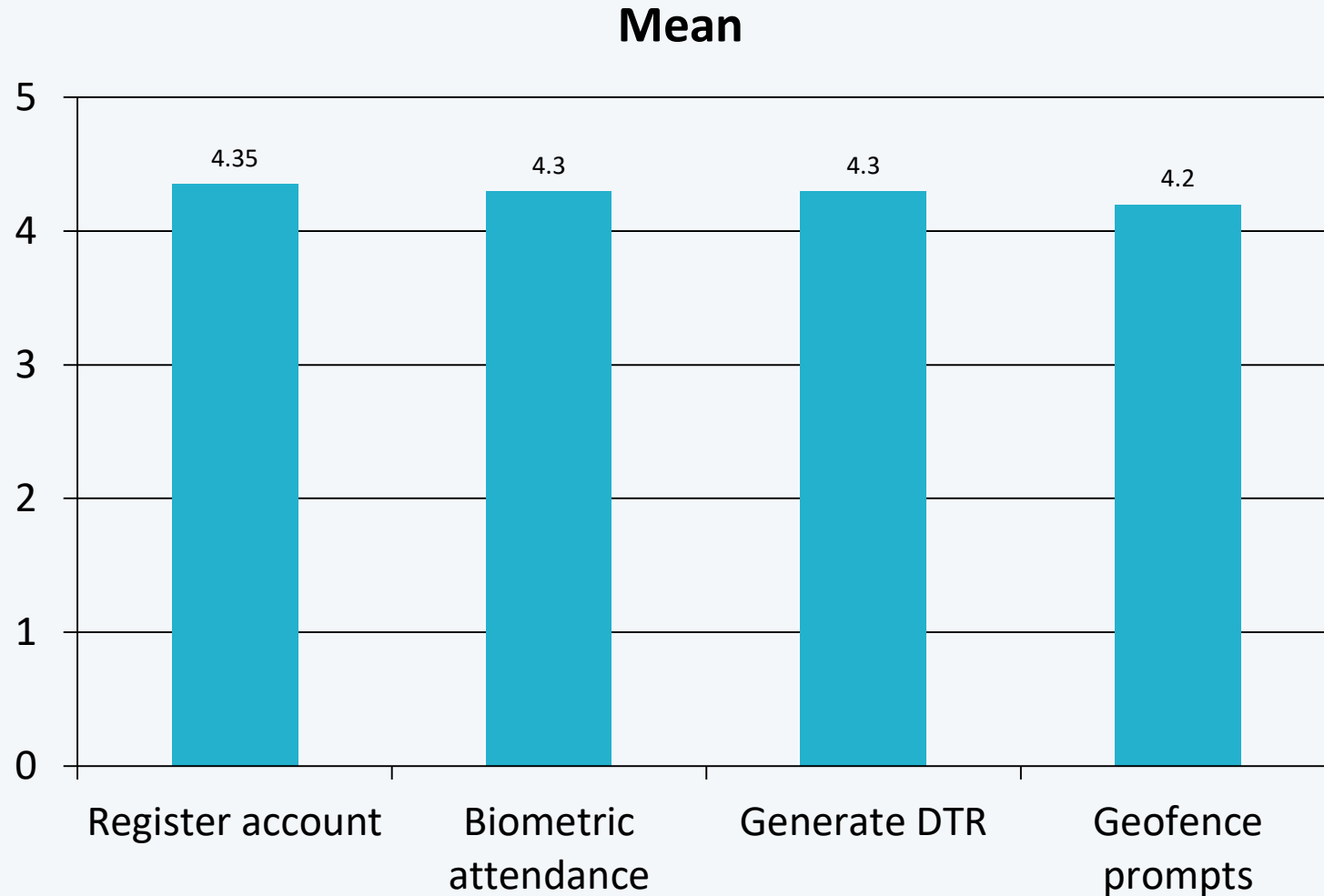
4.27

Overall mean

09

The User-Facing Features Also Rated Strongly

09



- Account registration: 4.35
- Biometric attendance: 4.30
- DTR generation: 4.30
- Geofence entry/exit prompts: 4.20
- All indicators were interpreted as "Agree."

A practical shift from “Who are you?” to “Are you here?”

4.27

Overall quality

5/5

Geofence validation

4.51

Usability

4.53

Portability

Next steps

Stronger registration verification • encryption/OTP/MFA • extra fingerprint verification • optional biometric alternative • total minutes late



Thank You

Mobile Biometrics Attendance Management System with Geofencing

